From: Bill Goetsch <bill@michagbusiness.net>
Sent: 11/24/2021 4:32:42 PM
To: "Doug Langworthy" <dlangworthy@Cadillac-MI.net>
Subject: Re: 2021 Biosolids PFAS Lab Report
Attachments: Cadillac PFOS notification_2021.pdf

Doug here is the PFOS notification we utilize. I filled in your information. Please confirm that the owners and growers have been notified so we can start first thing Monday.

Have a great Thanksgiving!!

Bill

On Wed, Nov 24, 2021 at 2:10 PM Doug Langworthy <dlangworthy@cadillac-mi.net> wrote:

Bill,
Attached is our PFAS results. I have spoke to the farmer and land owner. They are both waiting for my email. Thanks

Sent from my iPhone

Begin forwarded message:

From: Lab <lab@cadillac-mi.net>
Date: November 24, 2021 at 2:08:30 PM EST
To: Doug Langworthy <dlangworthy@cadillac-mi.net>
Subject: 2021 Biosolids PFAS Lab Report

As requested. Any questions, just send a reply.

Thanks,

Cindy A. Tomaszewski

Lab Supervisor

City of Cadillac – Utilities Dept.

231-775-2368 (lab)

231-878-6230 (cell)

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William Goetsch
Technical Services
616-232-0517
bill@michagbusiness.net
Michigan Agribusiness Solutions, LLC
3050 Freeway Lane
Saginaw, MI 48601
989-399-0800

ATTACHMENT NAME:

Cadillac PFOS notification_2021.pdf

ATTACHMENT TYPE:

Adobe Portable Document Format (PDF) compound image

Michigan AgriBusiness Solutions
3050 Freeway Lane - Saginaw,- MI - 48601 - Phone: (989) 399-0800 - Fax (989) 399-0811

Date: _______________

To all our Biosolids land application participants,

We at MAS would first like to thank you for you continued participation in our Biosolids recycling programs.

There has been much attention in Michigan to a class of chemicals called PFAS. The State of Michigan several years ago established the Michigan PFAS Action Response Team (MPART) over seen by the Michigan Department of Environment Great Lakes and Energy (EGLE) to address PFAS in Michigan. MPART has worked hard in identifying past and current sources of PFAS, and most importantly the limitation of PFAS release to the environment. Two forms of PFAS have not been in production for several years, PFOS & PFOA. Most of the news you hear about PFAS relate primarily to “water”, such as drinking water or surface waters. Of the hundreds of PFAS chemicals a few have shown to be found in Biosolids and one, PFOS, seems to be the most common. EGLE has addressed PFOS in Biosolids, through source reduction, with much success.

EGLE has required all wastewater treatment plants (WWTP) that land apply to test their Biosolids for PFAS. EGLE has established guidelines for the levels of PFAS and specifically PFOS for Biosolids to be land applied. Currently the ceiling concentration for PFOS is 150 parts per billion (ppb). We are providing landowners and farmers the level of PFOS most recently found in Biosolids that may be applied. This information is being shared so that Biosolids recycling participants are not only informed but hopefully appreciative of the scrutiny that the Biosolids being land applied are put through.

Following you will find a letter drafted by the Biosolids Team to give more information on PFAS and some information on the team working with PFAS and Biosolids. This letter should also have the most recent analytical data regarding PFOS in the Biosolids at the facility you participate with and also the contact number of the WWTP. Additionally, attached is a map of the EGLE personnel responsible for addressing any questions that you may have regarding PFAS and Biosolids.

We hope you have found this information informative. If you have questions of MAS please contact Bill Goetsch at 616-232-0517.

We look forward to working with you in the Future.

MAS

Biosolids and PFAS: Quick Facts for Landowners/Farmers

What are Biosolids? Biosolids are the treated materials produced during the processing of wastewater at a wastewater treatment plant (WWTP) (also known as a water resource recovery facility). Biosolids are rich in nutrients and organic matter and may be used as fertilizer or soil amendments (a beneficial use). A biosolids' quality and their proper use are regulated by the Michigan Department of Environment, Great Lakes, and Energy (EGLE) and the U.S. Environmental Protection Agency (USEPA). EGLE and the USEPA require biosolids to undergo a treatment process and be tested for certain pollutants to protect human health and the environment. Those processes refine the biosolids so that they can be applied at agronomic rates, providing a stable and valuable source of plant nutrients and soil structural enhancements.

What are PFAS and how do they get in wastewater? Per- and Polyfluoroalkyl substances (PFAS) are a large group of chemicals used for decades in industrial, commercial, and domestic settings and are found worldwide. Typical materials or processes that use or contain PFAS include firefighting foam, chrome plating, cookware coatings, waterproofing on clothing and carpet, and even food wrappers. Some PFAS, including Perflurooctane Sulfonate (PFOS), which are most commonly found in biosolids, have been phased out of production in the United States and are no longer approved for use. Even though they have not been used for years, their legacy remains.

WWTPs do not generate PFAS chemicals, though they may receive discharges from certain industrial or commercial sources who have used PFAS. As a result, PFAS may be found in treated wastewater and biosolids. Some of those PFAS are known to travel through water, can linger in the environment, and have the potential to impact the soil, water and crops. PFAS has been found to build up in the tissue of fish and deer in Michigan and in some areas led to consumption advisories. Studies by EGLE and USEPA are underway to determine the impact of PFAS on animals, animal products and crops. Some studies have shown that certain PFAS may harm human health, therefore, it is important to minimize exposure to all sources of PFAS in drinking water and food. For more information and referencing, please go to the website listed below.

Michigan's response to reducing PFAS in biosolids. Currently, USEPA is working to complete a risk-based evaluation of PFAS in biosolids. In the interim, EGLE's strategy is a deliberative, disciplined approach which focuses on identifying and reducing significant sources of PFAS entering WWTPs and preventing industrially impacted biosolids from being land applied. These efforts have helped WWTPs reduce PFAS concentrations in their biosolids.

The WWTP you are receiving biosolids from is:

Cadillac WWTP

You can reach the facility at:

Doug Langworthy 231-510-0238

Its most recent PFOS testing result is:

<5.3 ppb on 7/20/21

Presently EGLE's threshold concentration for PFOS in biosolids to be considered industrially impacted is 150 ppb.

In 2018, EGLE launched the Industrial Pretreatment Program (IPP) PFAS Initiative which has been successful in working with WWTPs to identify, reduce, and monitor sources of PFOS. Sixty-nine percent (69 percent) of the 95 WWTPs tested at the beginning of this initiative have already met water quality standards. Since the initial work in 2018, through aggressive source reduction efforts, the remaining facilities have continued their effective PFOS concentration reductions in treated wastewater by 49 to 99 percent compared to pre-2018 levels. For more information and referencing, please go to the website listed below.

Additionally, EGLE has established a protocol for WWTPs that may have biosolids impacted by PFAS industrial discharges. This includes a threshold concentration level (maximum of 150 parts per billion [ppb] for PFOS in biosolids) and monitoring requirements. Any WWTP that exceeds the threshold will not be allowed to land apply biosolids until they establish long-term measures for pre-treatment, eliminate their industrial sources of PFOS, and demonstrate that PFOS concentrations in their biosolids are consistently testing below the threshold concentration.

If you, your neighbors, or your customers have questions, you can find more information on PFAS in Biosolids by visiting the Michigan PFAS Action Response Team PFAS Land Application Workgroup web page Michigan.gov/PFASLandApplication. You may also reach out to the staff people listed on the [Statewide Biosolids and PFAS Contact map](https://www.michigan.gov/documents/egle/egle-mdard-biosolids-pfas-staff_720327_7.pdf) located at the following web page https://www.michigan.gov/documents/egle/egle-mdard-biosolids-pfas-staff_720327_7.pdf

Michigan Biosolids Team

For information or assistance on this publication, please contact the Biosolids Program through the EGLE Environmental Assistance Center at 800-662-9278. This publication is available in alternative formats upon request.

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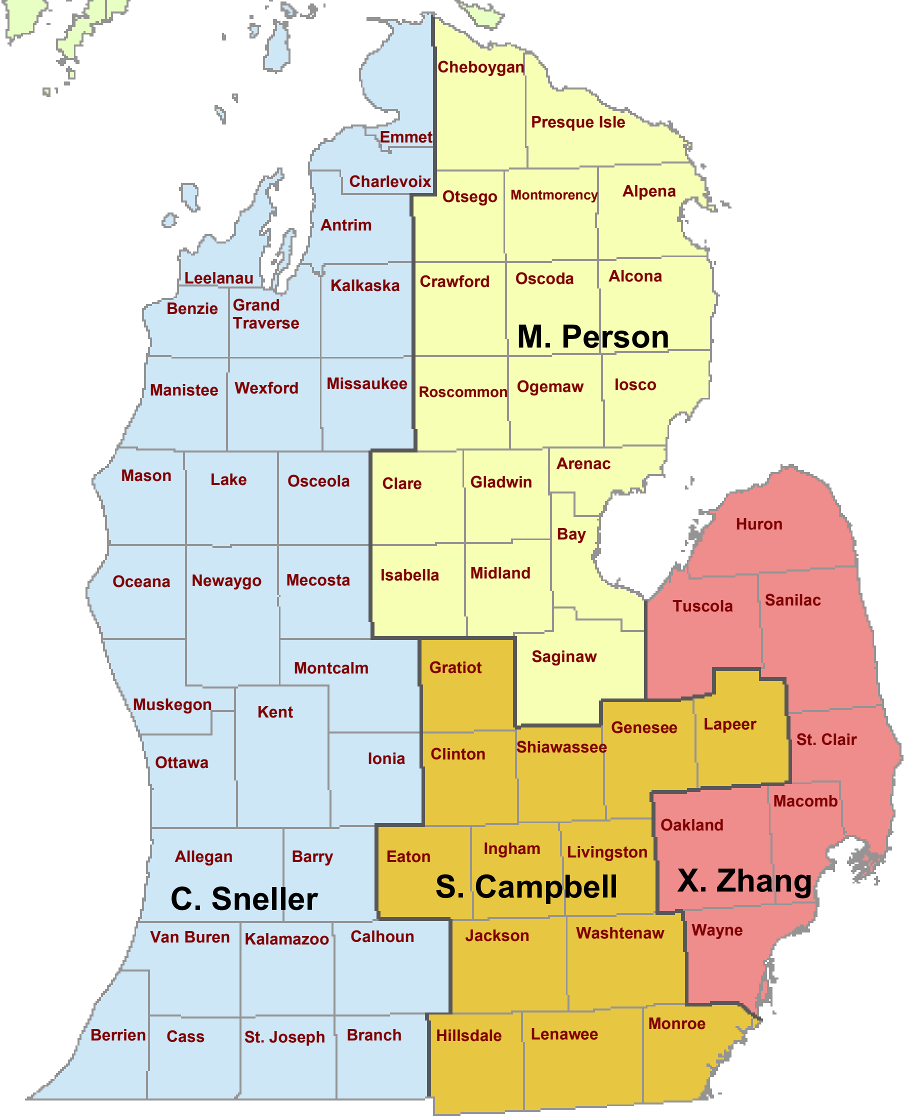
PFAS in Biosolids
State Contacts

Michigan.gov/Biosolids



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3/25/2021